

PAAVAI ENGINEERING COLLEGE

(An Autonomous Institution)
A NATIONAL LEVEL TECHNICAL SYMPOSIUM

TECHFINIX'26
PROJECT EXPO

Post-Quantum Secure Communication Platform

OBJECTIVE:

- The main aim of this project is to protect digital communication from future quantum computer attacks.
- The focus of this project is to eliminate "Harvest Now, Decrypt Later" data theft in chat, mail, and file storage.

ABSTRACT:

- Digital communication platforms become essential for everyday data exchange.
- The main drawback of current systems (RSA & ECC) is that they can be easily broken by quantum computers.
- The proposed system aims to provide quantum-safe encryption using NIST-standardized Post-Quantum Cryptography (PQC).

INNOVATION:

- The post-quantum cryptography concept is a recent development. We are enhancing the security by implementing true dual-PQC (both Quantum Key Exchange and Quantum Digital Signatures) for real-time chat, secure mail, and file vault.

TECHNOLOGY INVOLVED:

- **Post-Quantum KEM:** In this project we have used ML-KEM-768 for quantum-safe key exchange.
- **Post-Quantum Digital Signature:** ML-DSA-65 is used for sender authentication and message signing.
- **Symmetric Encryption:** AES-256-GCM authenticated cipher with 512-bit SHA3 integrity check.

Flask-SocketIO WebSockets
liboqs Native C Library
Three.js 3D WebGL Interface

OUTCOMES:

- 100% Quantum-Proof and enhanced communication security.
- The total plaintext leakage is zero; no sensitive data is stored in plain text.
- The cryptographic speed and reliability is very high.

[Image 1] 3D Quantum Portal	[Image 2] Encrypted Live Chat	[Image 3] Encrypted File Vault
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Project By-

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Guided by-

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